

Student Evaluation Portal Using C

1. Introduction

The Student Evaluation Portal is a C-based system designed to manage and evaluate student academic performance efficiently. It enables teachers to upload student marks and attendance securely, while students can log in using institutional email credentials to view their results along with automated suggestions. This project reflects a basic implementation of a real-world academic system.

2. Objectives

The main objectives of this project are:

- To develop a secure academic portal using C programming
- To implement authentication using institutional email
- To manage student marks and attendance data
- To generate automated performance-based suggestions
- To understand real-world system development

3. Features

- Teacher Registration & Login
- Student Registration & Login
- DIU Email Validation System
- Student Data Upload (Marks & Attendance)
- Result Viewing System
- Automated Suggestion System
- Error Handling for unavailable data

4. Tools & Technologies

- Programming Language: C
- Core Concepts:
 - ☐ Structures
 - ☐ Functions
 - ☐ Conditional Statements
 - ☐ String Handling
- Platform: Console-based application

5. System Design

- The system follows a modular design approach:
- Authentication Module: Handles teacher and student login/registration

- Data Management Module: Stores student information using structures
- Evaluation Module: Analyzes marks and attendance
- Output Module: Displays results and suggestions
- This modular design ensures better organization and maintainability.

Flow of the system:

- Start the program
- Display menu options
- Teacher registers/logs in
- Teacher uploads student data
- Student registers/logs in
- Student enters ID
- System searches for data
- If found → display result and suggestion
- If not found → show warning message
- Exit or repeat process

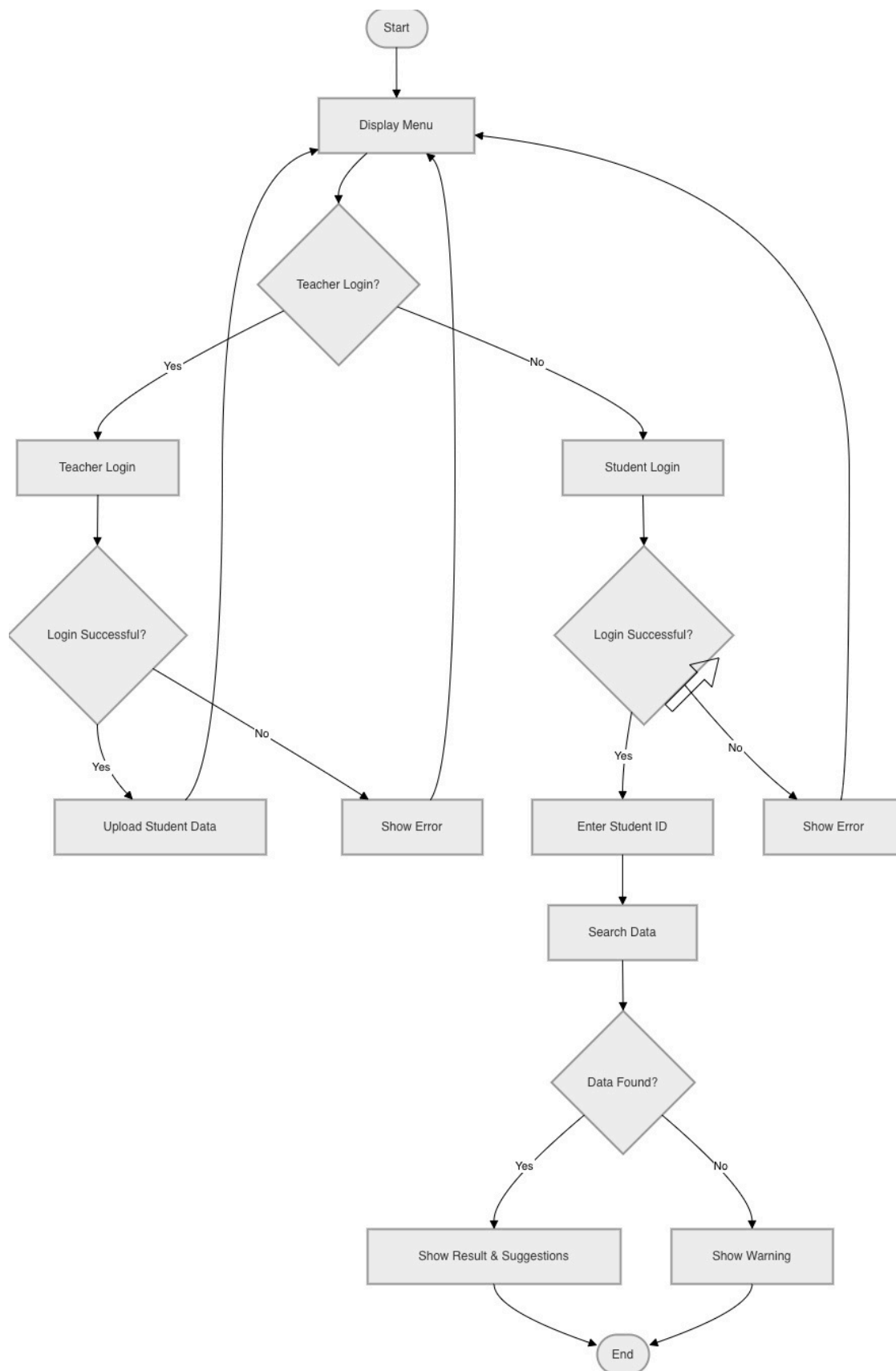


Figure 1: Flowchart of Student Evaluation Portal

6. Implementation

The system is implemented using structured programming in C. Student data is stored using arrays of structures. Authentication is handled by comparing user input with stored credentials. Email validation is performed using string matching functions. Conditional logic is used to generate suggestions based on marks and attendance.

7. Sample Input/Output

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main.c

```
1 #include <stdio.h>
2 #include <string.h>
3
4 #define MAX 50
5
6 struct Student {
7     char id[100];
8     float marks;
9     float attendance;
10 };
11
12 struct Student s[MAX];
13 int n = 0;
14
15 int isValidDIUEmail(char email[]) {
16     char *domain = "@diu.edu.bd";
17     if(strstr(email, domain) != NULL)
18         return 1;
19     return 0;
20 }
21
22 char teacherEmail[100];
23 int teacherPass;
24
25 void teacherRegister() {
26     printf("\n--- Teacher Registration ---\n");
27     printf("Enter Email: ");
28     scanf("%s", teacherEmail);
29
30     if(!isValidDIUEmail(teacherEmail)) {
31         printf("Invalid Email! Must end with @diu.edu.bd\n");
32         return;
33     }
34
35     printf("Enter 4-digit Password: ");
36     scanf("%d", &teacherPass);
```

```
1. Teacher Register
2. Teacher Login & Upload
3. Student Register
4. Student Login
5. Exit
Choice:
```

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22 char teacherEmail[100];
23 int teacherPass;
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25 void teacherRegister() {
26     printf("\n--- Teacher Registration ---\n");
27     printf("Enter Email: ");
28     scanf("%s", teacherEmail);
29
30     if(!isValidDIUEmail(teacherEmail)) {
31         printf("Invalid Email! Must end with @diu.edu.bd\n");
32         return;
33     }
34
35     printf("Enter 4-digit Password: ");
36     scanf("%d", &teacherPass);
```

```
1. Teacher Register
2. Teacher Login & Upload
3. Student Register
4. Student Login
5. Exit
Choice: 3

--- Student Registration ---
Enter Email: shad@diu.edu.bd
Enter 4-digit Password: 2341
Registration Done!

1. Teacher Register
2. Teacher Login & Upload
3. Student Register
4. Student Login
5. Exit
Choice: 
```

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15 int isValidDIUEmail(char email[]) {
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17     if(strstr(email, domain) != NULL)
18         return 1;
19     return 0;
20 }
21
22 char teacherEmail[100];
23 int teacherPass;
24
25 void teacherRegister() {
26     printf("\n--- Teacher Registration ---\n");
27     printf("Enter Email: ");
28     scanf("%s", teacherEmail);
29
30     if(!isValidDIUEmail(teacherEmail)) {
31         printf("Invalid Email! Must end with @diu.edu.bd\n");
32         return;
33     }
34
35     printf("Enter 4-digit Password: ");
36     scanf("%d", &teacherPass);
```

Input

1. Teacher Register
2. Teacher Login & Upload
3. Student Register
4. Student Login
5. Exit
Choice: 1

--- Teacher Registration ---
Enter Email: setu@diu.edu.bd
Enter 4-digit Password: 2413
Registration Done!

1. Teacher Register
2. Teacher Login & Upload
3. Student Register
4. Student Login
5. Exit
Choice: 2

--- Teacher Login ---
Email: setu@diu.edu.bd
Password: 2413
Login Successful!

Enter number of students: 2

Student 1 ID: 251-35-280
Marks: 87
Attendance: 90

Student 2 ID: 251-35-039
Marks: 45
Attendance: 50

Data Uploaded Successfully!

1. Teacher Register
2. Teacher Login & Upload
3. Student Register
4. Student Login
5. Exit
Choice:

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Run Debug Stop Share Save {} Beautify Language C

main.c

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25 void teacherRegister() {
26     printf("\n--- Teacher Registration ---\n");
27     printf("Enter Email: ");
28     scanf("%s", teacherEmail);
29
30     if(!isValidDIUEmail(teacherEmail)) {
31         printf("Invalid Email! Must end with @diu.edu.bd\n");
32         return;
33     }
34
35     printf("Enter 4-digit Password: ");
36     scanf("%d", &teacherPass);
```

Input

Enter Email: shad@diu.edu.bd
Enter 4-digit Password: 2431
Registration Done!

1. Teacher Register
2. Teacher Login & Upload
3. Student Register
4. Student Login
5. Exit
Choice: 4

--- Student Login ---
Email: shad@diu.edu.bd
Password: 2431
Login Successful!

Enter Your Student ID: 251-35-280

===== RESULT =====
Marks: 87.00
Attendance: 90.00%
Suggestion:
- Good performance, keep it up!

1. Teacher Register
2. Teacher Login & Upload
3. Student Register
4. Student Login
5. Exit
Choice: 4

--- Student Login ---
Email: shad@diu.edu.bd
Password: 2431
Login Successful!

Enter Your Student ID: 251-35-210

b Teacher hasn't uploaded your data yet. Please wait.

1. Teacher Register
2. Teacher Login & Upload
3. Student Register
4. Student Login
5. Exit
Choice:

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LanguageC

Input

main.c

```
1 #include <stdio.h>
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32         return;
33     }
34
35     printf("Enter 4-digit Password: ");
36     scanf("%d", &teacherPass);
```

Attendance: 90

Student 2 ID: 251-35-039

Marks: 45

Attendance: 50

Data Uploaded Successfully!

1. Teacher Register

2. Teacher Login & Upload

3. Student Register

4. Student Login

5. Exit

Choice: 3

--- Student Registration ---

Enter Email: shad@diu.edu.bd

Enter 4-digit Password: 2431

Registration Done!

1. Teacher Register

2. Teacher Login & Upload

3. Student Register

4. Student Login

5. Exit

Choice: 4

--- Student Login ---

Email: shad@diu.edu.bd

Password: 2431

Login Successful!

Enter Your Student ID: 251-35-280

===== RESULT =====

Marks: 87.00

Attendance: 90.00%

Suggestion:

- Good performance, keep it up!

1. Teacher Register

2. Teacher Login & Upload

3. Student Register

4. Student Login

5. Exit

Choice:

8. Ethical Theories Applied

The system evaluates decisions based on:

- Ensuring secure login using institutional email
- Preventing unauthorized access to student data
- Maintaining fairness in evaluation
- Protecting basic data privacy

9. Conclusion

The Student Evaluation Portal demonstrates how fundamental programming concepts can be applied to build a simple yet functional academic system. It ensures structured data handling and secure access while providing useful feedback to students.

10. Future Improvements

- File handling for permanent data storage
- Graphical User Interface (GUI)
- Support for multiple users
- Database integration
- Advanced performance analytics

11. Author

Name: Md Tasnimul Islam Shad
Project Title: Student Evaluation Portal
Programming Language: C

12. Learning Outcomes / Course Integration

This project is primarily influenced by knowledge gained from the following courses:

Microsoft → Data Structure & Algorithm

IBM → Introduction to Software Engineering

Google → Git & GitHub + Project Management

These courses played a significant role in shaping the system design, structured implementation, and overall development process of this project.

This project shows how basic knowledge from different subjects can be combined to create a simple and useful application.